

Documento comprobatorio y explicativo

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The screenshot displays a C++ development environment with three main panels:

- Left Panel (Code Editor):** Shows the `intermediary.cpp` file. The code includes headers for `<sstream>`, `<string>`, and `<chrono>`. It defines a `POST` function that takes a `figura` string and a `LISTAR` function that takes a `Salir` string. The `main` function calls `POST` and `LISTAR` with the input `figura: Cat` and `Salir: Lion`. The output shows the response for `figura: Cat` as `5 brick 2x2 yellow`, `4 brick 2x4 white`, `2 brick 2x4 yellow`, and `1 brick 2x6 yellow`.
- Middle Panel (Terminal):** Shows the execution of the program. It displays the output of the `POST` and `LISTAR` functions, followed by the response for `figura: Lion` as `5 brick 2x2 yellow`, `4 brick 2x4 white`, `2 brick 2x4 yellow`, and `1 brick 2x6 yellow`.
- Right Panel (File Explorer):** Shows the file structure of the project, including `intermediary.cpp`, `intermediary.h`, `main.cpp`, and `main.h`.

The code in `intermediary.cpp` is as follows:

```
#include <sstream>
#include <string>
#include <chrono>

using namespace std;

// ... (code for POST and LISTAR functions) ...

int main() {
    string figura;
    string Salir;

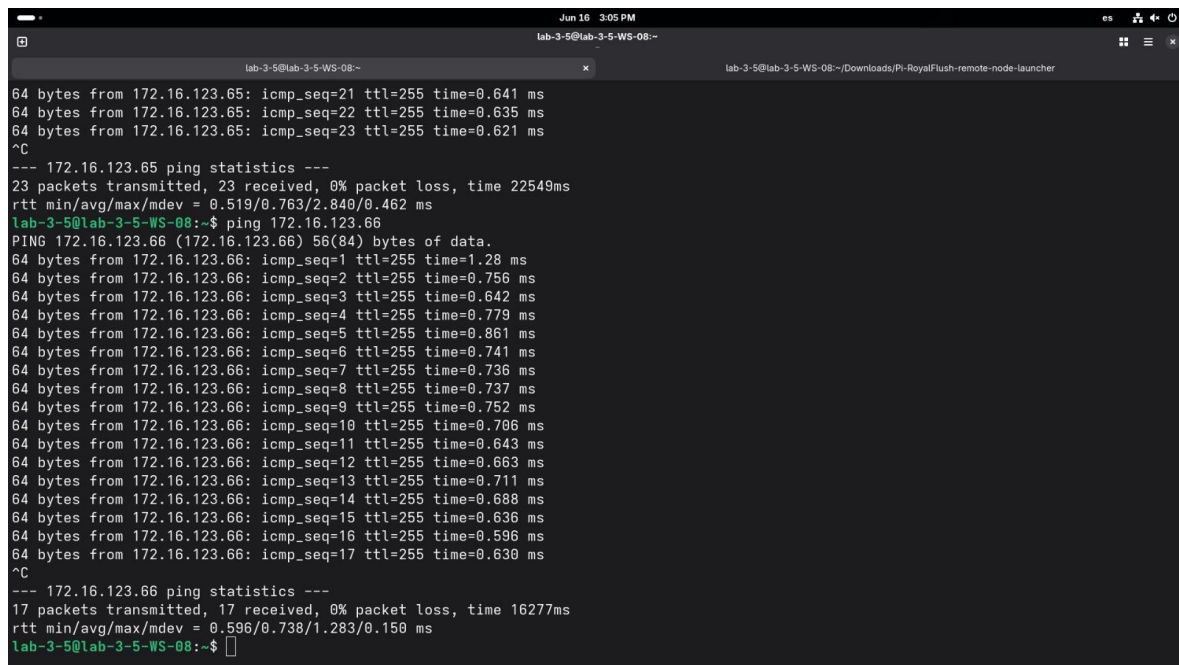
    POST(figura);
    LISTAR(Salir);

    return 0;
}
```

En esta primera imagen vemos como la comunicación local funciona, primero se levanta el servidor, actualmente en el puerto 1235, esperando a tener conexión con el intermediario. Las Raw request (P/R/) que se pueden ver en el servidor son prueba de que el intermediario encontró el servidor local, iniciándose con su ip de red y el puerto respectivo como argumentos, Con esto el intermediario actualiza en su routemap el valor con la key [LOCAL] y se obtiene acceso desde el cliente que en este caso usamos NachOS pero se va a volver al cliente inicial dado que ya no es necesario NachOS.

Nota: A su misma vez el intermediario está buscando forks de otras islas, pero como no habíamos activado la comunicación directa en las broadcast de cada vlan desde el L3 no encontraba aún. Luego de este cambio se logró establecer la conexión pero

seguía sin haber comunicación, debido a que algunos hicimos conexión con UDP y comunicación con TCP. En la clase del viernes como grupo decidimos cambiar todos a conexión y comunicación por UDP sobre el puerto 2026, usando el mismo hilo para esto, lo cual nos va a solucionar estos problemas, por lo tanto este cambio es lo que está pendiente y volver a hacer las pruebas de comunicación con otros forks para ir quedando listos con la etapa 4.



```
Jun 16 3:05 PM
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.65
64 bytes from 172.16.123.65: icmp_seq=21 ttl=255 time=0.641 ms
64 bytes from 172.16.123.65: icmp_seq=22 ttl=255 time=0.635 ms
64 bytes from 172.16.123.65: icmp_seq=23 ttl=255 time=0.621 ms
^C
--- 172.16.123.65 ping statistics ---
23 packets transmitted, 23 received, 0% packet loss, time 22549ms
rtt min/avg/max/mdev = 0.519/0.763/2.840/0.462 ms
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.66
PING 172.16.123.66 (172.16.123.66) 56(84) bytes of data.
64 bytes from 172.16.123.66: icmp_seq=1 ttl=255 time=1.28 ms
64 bytes from 172.16.123.66: icmp_seq=2 ttl=255 time=0.756 ms
64 bytes from 172.16.123.66: icmp_seq=3 ttl=255 time=0.642 ms
64 bytes from 172.16.123.66: icmp_seq=4 ttl=255 time=0.779 ms
64 bytes from 172.16.123.66: icmp_seq=5 ttl=255 time=0.861 ms
64 bytes from 172.16.123.66: icmp_seq=6 ttl=255 time=0.741 ms
64 bytes from 172.16.123.66: icmp_seq=7 ttl=255 time=0.736 ms
64 bytes from 172.16.123.66: icmp_seq=8 ttl=255 time=0.737 ms
64 bytes from 172.16.123.66: icmp_seq=9 ttl=255 time=0.752 ms
64 bytes from 172.16.123.66: icmp_seq=10 ttl=255 time=0.706 ms
64 bytes from 172.16.123.66: icmp_seq=11 ttl=255 time=0.643 ms
64 bytes from 172.16.123.66: icmp_seq=12 ttl=255 time=0.663 ms
64 bytes from 172.16.123.66: icmp_seq=13 ttl=255 time=0.711 ms
64 bytes from 172.16.123.66: icmp_seq=14 ttl=255 time=0.688 ms
64 bytes from 172.16.123.66: icmp_seq=15 ttl=255 time=0.636 ms
64 bytes from 172.16.123.66: icmp_seq=16 ttl=255 time=0.596 ms
64 bytes from 172.16.123.66: icmp_seq=17 ttl=255 time=0.630 ms
^C
--- 172.16.123.66 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16277ms
rtt min/avg/max/mdev = 0.596/0.738/1.283/0.150 ms
lab-3-5@lab-3-5-WS-08:~$
```

En esta imagen comprobamos la conexión del host de pruebas con el L2 local de Brigada A, ip .66, el cuál usa esta config:

L2-Isla4#show running-config

Building configuration...

Current configuration : 3343 bytes

!

version 12.2

no service pad

service timestamps debug uptime

service timestamps log uptime

no service password-encryption

```
!  
hostname L2-Isla4  
!  
!  
no aaa new-model  
system mtu routing 1500  
ip subnet-zero  
ip dhcp excluded-address 172.16.123.65  
ip dhcp excluded-address 172.16.123.66  
!  
ip dhcp pool ISLA4  
    network 172.16.123.64 255.255.255.240  
    default-router 172.16.123.65  
    dns-server 8.8.8.8  
!  
!  
!  
!  
no file verify auto  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
vlan internal allocation policy ascending  
!  
interface FastEthernet0/1  
    switchport access vlan 340  
    switchport mode access
```

spanning-tree portfast

!

interface FastEthernet0/2

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/3

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/4

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/5

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/6

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

```
interface FastEthernet0/7
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
```

```
interface FastEthernet0/8
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
```

```
interface FastEthernet0/9
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
```

```
interface FastEthernet0/10
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
```

```
interface FastEthernet0/11
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
```

```
interface FastEthernet0/12
switchport access vlan 340
```

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/13

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/14

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/15

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/16

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/17

switchport access vlan 340

switchport mode access

spanning-tree portfast

!

```
interface FastEthernet0/18
switchport access vlan 340
switchport mode access
spanning-tree portfast
```

!

```
interface FastEthernet0/19
switchport access vlan 340
switchport mode access
spanning-tree portfast
```

!

```
interface FastEthernet0/20
switchport access vlan 340
switchport mode access
spanning-tree portfast
```

!

```
interface FastEthernet0/21
switchport access vlan 340
switchport mode access
spanning-tree portfast
```

!

```
interface FastEthernet0/22
switchport access vlan 340
switchport mode access
spanning-tree portfast
```

!

```
interface FastEthernet0/23
```

```
switchport access vlan 340
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 340
switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
no ip route-cache
!
interface Vlan340
ip address 172.16.123.66 255.255.255.240
no ip route-cache
!
ip default-gateway 172.16.123.65
ip http server
!
control-plane
!
!
```

line con 0

End

```

lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.65
PING 172.16.123.65 (172.16.123.65) 56(84) bytes of data:
64 bytes from 172.16.123.65: icmp_seq=1 ttl=255 time=0.647 ms
64 bytes from 172.16.123.65: icmp_seq=2 ttl=255 time=0.630 ms
64 bytes from 172.16.123.65: icmp_seq=3 ttl=255 time=0.653 ms
64 bytes from 172.16.123.65: icmp_seq=4 ttl=255 time=0.698 ms
64 bytes from 172.16.123.65: icmp_seq=5 ttl=255 time=0.651 ms
64 bytes from 172.16.123.65: icmp_seq=6 ttl=255 time=0.667 ms
64 bytes from 172.16.123.65: icmp_seq=7 ttl=255 time=0.649 ms
64 bytes from 172.16.123.65: icmp_seq=8 ttl=255 time=0.658 ms
64 bytes from 172.16.123.65: icmp_seq=9 ttl=255 time=0.628 ms
64 bytes from 172.16.123.65: icmp_seq=10 ttl=255 time=0.657 ms
64 bytes from 172.16.123.65: icmp_seq=11 ttl=255 time=0.635 ms
64 bytes from 172.16.123.65: icmp_seq=12 ttl=255 time=0.678 ms
64 bytes from 172.16.123.65: icmp_seq=13 ttl=255 time=0.625 ms
64 bytes from 172.16.123.65: icmp_seq=14 ttl=255 time=0.631 ms
64 bytes from 172.16.123.65: icmp_seq=15 ttl=255 time=1.25 ms
64 bytes from 172.16.123.65: icmp_seq=16 ttl=255 time=0.790 ms
64 bytes from 172.16.123.65: icmp_seq=17 ttl=255 time=2.84 ms
64 bytes from 172.16.123.65: icmp_seq=18 ttl=255 time=0.519 ms
64 bytes from 172.16.123.65: icmp_seq=19 ttl=255 time=0.527 ms
64 bytes from 172.16.123.65: icmp_seq=20 ttl=255 time=0.624 ms
64 bytes from 172.16.123.65: icmp_seq=21 ttl=255 time=0.641 ms
64 bytes from 172.16.123.65: icmp_seq=22 ttl=255 time=0.635 ms
64 bytes from 172.16.123.65: icmp_seq=23 ttl=255 time=0.621 ms
^C
--- 172.16.123.65 ping statistics ---
23 packets transmitted, 23 received, 0% packet loss, time 22549ms
rtt min/avg/max/mdev = 0.519/0.763/2.840/0.462 ms
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.66
PING 172.16.123.66 (172.16.123.66) 56(84) bytes of data.

```

Aquí apreciamos la conexión exitosa del host de pruebas con el L3 del grupo, ip .65.

```
Jun 16 3:06 PM
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.66
64 bytes from 172.16.123.66: icmp_seq=11 ttl=255 time=0.643 ms
64 bytes from 172.16.123.66: icmp_seq=12 ttl=255 time=0.663 ms
64 bytes from 172.16.123.66: icmp_seq=13 ttl=255 time=0.711 ms
64 bytes from 172.16.123.66: icmp_seq=14 ttl=255 time=0.688 ms
64 bytes from 172.16.123.66: icmp_seq=15 ttl=255 time=0.636 ms
64 bytes from 172.16.123.66: icmp_seq=16 ttl=255 time=0.596 ms
64 bytes from 172.16.123.66: icmp_seq=17 ttl=255 time=0.630 ms
^C
--- 172.16.123.66 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16277ms
rtt min/avg/max/mdev = 0.596/0.738/1.283/0.150 ms
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.34
PING 172.16.123.34 (172.16.123.34) 56(84) bytes of data.
64 bytes from 172.16.123.34: icmp_seq=1 ttl=64 time=0.785 ms
64 bytes from 172.16.123.34: icmp_seq=2 ttl=64 time=0.634 ms
64 bytes from 172.16.123.34: icmp_seq=3 ttl=64 time=0.462 ms
64 bytes from 172.16.123.34: icmp_seq=4 ttl=64 time=0.649 ms
64 bytes from 172.16.123.34: icmp_seq=5 ttl=64 time=0.581 ms
64 bytes from 172.16.123.34: icmp_seq=6 ttl=64 time=0.627 ms
64 bytes from 172.16.123.34: icmp_seq=7 ttl=64 time=0.563 ms
64 bytes from 172.16.123.34: icmp_seq=8 ttl=64 time=0.468 ms
64 bytes from 172.16.123.34: icmp_seq=9 ttl=64 time=0.583 ms
64 bytes from 172.16.123.34: icmp_seq=10 ttl=64 time=0.537 ms
64 bytes from 172.16.123.34: icmp_seq=11 ttl=64 time=0.513 ms
64 bytes from 172.16.123.34: icmp_seq=12 ttl=64 time=0.524 ms
64 bytes from 172.16.123.34: icmp_seq=13 ttl=64 time=0.549 ms
^C
--- 172.16.123.34 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12324ms
rtt min/avg/max/mdev = 0.462/0.575/0.785/0.082 ms
lab-3-5@lab-3-5-WS-08:~$
```

Aquí vemos la conexión exitosa con un L2 externo, el de la isla 3, en algunas especificaciones sale como isla 2 pero la cuestión es que es la ip .34, grupo de Anferny, Paul, etc.

```
Jun 16 3:08 PM
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.35
64 bytes from 172.16.123.35: icmp_seq=8 ttl=64 time=0.468 ms
64 bytes from 172.16.123.35: icmp_seq=9 ttl=64 time=0.583 ms
64 bytes from 172.16.123.35: icmp_seq=10 ttl=64 time=0.537 ms
64 bytes from 172.16.123.35: icmp_seq=11 ttl=64 time=0.513 ms
64 bytes from 172.16.123.35: icmp_seq=12 ttl=64 time=0.524 ms
64 bytes from 172.16.123.35: icmp_seq=13 ttl=64 time=0.549 ms
^C
--- 172.16.123.35 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12324ms
rtt min/avg/max/mdev = 0.462/0.575/0.785/0.082 ms
lab-3-5@lab-3-5-WS-08:~$ ping 172.16.123.35
PING 172.16.123.35 (172.16.123.35) 56(84) bytes of data.
64 bytes from 172.16.123.35: icmp_seq=1 ttl=63 time=0.542 ms
64 bytes from 172.16.123.35: icmp_seq=2 ttl=63 time=0.638 ms
64 bytes from 172.16.123.35: icmp_seq=3 ttl=63 time=0.697 ms
64 bytes from 172.16.123.35: icmp_seq=4 ttl=63 time=0.662 ms
64 bytes from 172.16.123.35: icmp_seq=5 ttl=63 time=0.496 ms
64 bytes from 172.16.123.35: icmp_seq=6 ttl=63 time=0.508 ms
64 bytes from 172.16.123.35: icmp_seq=7 ttl=63 time=0.647 ms
64 bytes from 172.16.123.35: icmp_seq=8 ttl=63 time=0.496 ms
64 bytes from 172.16.123.35: icmp_seq=9 ttl=63 time=0.696 ms
64 bytes from 172.16.123.35: icmp_seq=10 ttl=63 time=0.509 ms
64 bytes from 172.16.123.35: icmp_seq=11 ttl=63 time=0.449 ms
64 bytes from 172.16.123.35: icmp_seq=12 ttl=63 time=0.523 ms
64 bytes from 172.16.123.35: icmp_seq=13 ttl=63 time=0.518 ms
64 bytes from 172.16.123.35: icmp_seq=14 ttl=63 time=1.11 ms
^C
--- 172.16.123.35 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13297ms
rtt min/avg/max/mdev = 0.449/0.606/1.111/0.161 ms
lab-3-5@lab-3-5-WS-08:~$
```

Por último, tenemos la prueba de conexión con un host de una isla externa, en este caso nuevamente la isla 3 con la ip .35

En resumen, las conexiones son exitosas en este punto, la comunicación local también y solo falta ajustar toda la comunicación externa a UDP para completar la etapa 4 y poder hacer solicitudes a otros forks.